

CHETANYA SINGH

SOFTWARE ENGINEER (AI/ML)

CONTACT

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- [LinkedIn Profile](#)
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PROFILE SUMMARY

B.Tech CSE-AI/ML student with hands-on experience engineering end-to-end artificial intelligence pipelines and scalable full-stack software systems. Proven track record developing responsive SaaS platforms using React, Flask, and horizontally scalable MySQL databases, alongside low-latency computer vision models (YOLO, PyTorch, TensorFlow). Passionate about applying supervised learning, probabilistic modeling, and data-driven software architecture to solve complex real-world problems. Eager to deepen expertise under industry leaders and contribute to impactful AI and software solutions.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY - AP 2024-2028

- B.Tech in CSE-AI/ML
- CGPA: 9.11/10

PODDAR BRIO INT. SCHOOL, BADLAPUR (XII) 2024

- Percentage: 90.8%

RELIANCE FOUNDATION SCHOOL, RAIGARD (X) 2022

- Percentage: 95.8%

SKILLS

Technical Skills

- Python, Java, C, React, Tailwind CSS, Flask, MySQL, R Programming
- Machine Learning Tools: Scikit-learn, OpenCV, TensorFlow, Keras, PyTorch, MobileNetV2, YOLO
- Data Libraries: Pandas, NumPy
- Visualization: Matplotlib, Seaborn
- Core CS/Mathematics: DSA, OOP, Linear Algebra, Probability

Soft Skills

- Analytical Problem Solving
- Agile Adaptability
- Project Execution
- Technical Leadership

PROJECT & LEADERSHIP EXPERIENCE

Real-Time Cognitive Fatigue and Drowsiness Monitor 2026

Academic & Personal Project

- Engineered an end-to-end computer vision pipeline using OpenCV and Haar Cascades for real-time facial isolation and eye-state tracking.
- Trained and optimized a MobileNetV2 Convolutional Neural Network (CNN) using TensorFlow/Keras, achieving >94% classification accuracy for alert versus fatigued states while optimizing for low-latency CPU performance.

Hostel/PG Management SaaS Platform 2025

Academic Project

- Built a responsive full-stack accommodation management platform using React and Tailwind CSS, powered by a stateless Python/Flask backend.
- Architected a horizontally scalable MySQL database via PyMySQL, capable of supporting thousands of concurrent users and processing millions of resident records without latency.

Military Aircraft Detection Model 2026

Personal Project

- Developed a real-time object detection pipeline using PyTorch and YOLO architectures to identify and classify military aircraft across diverse aerial and terrestrial backgrounds.
- Conducted data preprocessing, augmentation, and hyperparameter tuning on multi-class image datasets, achieving high mean Average Precision (mAP) while optimizing inference speed for edge deployment.

LANGUAGES

- English
- Hindi
- Marathi

CERTIFICATION

- Data Structures and Algorithms (DSA) - GeeksforGeeks